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Crystal Structures of Fluorinated Pyridines from Geometrical and Energetic Perspectives

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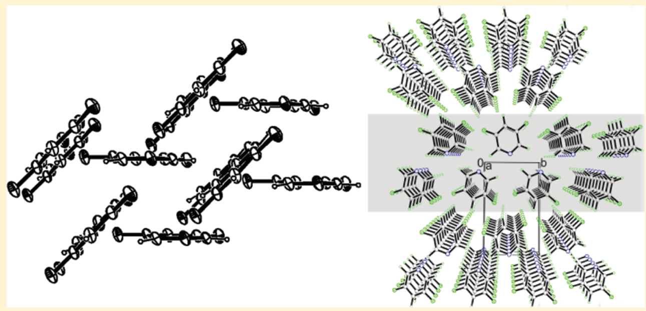
Supporting Information



ABSTRACT: The low melting compounds 3-fluoropyridine, 3,5-difluoropyridine, 2,3,5-trifluoropyridine, and 2,3,5,6-tetrafluoropyridine were crystallized in situ on the diffractometer and analyzed by X-ray diffraction. The comparison with pentafluoropyridine shows a consecutive dependence of the arrangement of pyridine molecules with the increasing fluorine substitution. Starting with the monosubstituted representative, the crystal packing changes stepwise from herringbone packing to parallel arrangement of the molecules in the trifluorinated pyridine and then switches back to the edge-to-face form by the perfluorinated compound. To get a deeper insight into the aggregation behavior of the flourinated pyridines, the crystal packing motives were also analyzed on the basis of ab initio quantum-chemical calculations of the intermolecular interaction energy, using the MP2/6-311G(d,p) method. This approach allows the indication of energetically rich fragments in the crystal structures, called basic structural motifs, and the very weak attractive or rather repulsive nature of F···F interactions.

INTRODUCTION

In recent years, halogen−halogen interactions involving C−Cl, C−Br, and C−I groups have become an important tool in crystal engineering.[1] This is caused by the character of interactions between these halogen atoms and the electron-donating atoms of the neighboring molecules. The existence of the so-called σ-hole surrounded by the region of a negative electrostatic potential creates a possibility for the formation of intermolecular halogen interactions.[2,3] The understanding of the possible modes of intermolecular interactions with participation of the chlorine, bromine, or iodine atoms provides suitable background for the prediction of the arrangement of neighboring molecules in the crystalline phase. For example, if a molecule contains an iodine atom and a carbonyl group, it is possible to expect the formation of an I···O halogen interaction causing a relevant orientation of the interacting molecules in the crystal.[4−6]

In contrast to halogen interactions involving C−Cl, C−Br, and C−I groups, the special aggregation behavior of fluorinated arenes is of high interest. A number of studies have been carried out that show that fluorine generates different types of packing motives via C−H···F, C−F···F, and C−F···π interactions, especially in the absence of strong hydrogen-bond donors and/or acceptors such as N−H, O−H, C═O, etc.[7−11] The question of whether F···F contacts can really be called weak attractive interactions or if they are simply caused by the close packing in the crystal is controversial. The nature of F···F interactions is still not fully understood because from the supramolecular perspective fluorine is a complex element.[12] According to Pauling,[13] it is expected that hydrogen bonds are preferably formed with electronegative atoms. While the F-anion is an excellent acceptor,[14] the organic C−F groups form only weak interactions as compared to typical H-bond acceptors such as oxygen and nitrogen.[15]

Nevertheless, there are a few compounds where F···F contacts were studied by a combination of NMR spectroscopy and AIM (Atoms in Molecules) calculations.[16] Several studies showed an increase of the frequency of F···F contacts with the increasing fluorination degree in the molecule.[17−19]

Deya et al. analyzed F···F interactions in some fluorouracil derivatives computationally using ab initio calculations and Bader's "Atoms in Molecules" theory and reported about a stabilizing effect of F···F contacts because they belong to the type II of halogen−halogen interactions.[20] Type-II interactions may be understood according to a model of halogen bonds that

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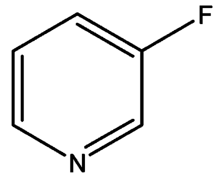
assigns a positive polarization in the polar region of the halogen atom and a negative polarization in its equatorial region.[21−23]

Such uncertainty in the character of intermolecular interactions with fluorine atoms involved causes a poor understanding of the crystal structures of fluorinated compounds. It is difficult to explain why molecules are arranged in a specific manner in the crystal and which interactions are dominating the crystal packing. This requires a systematic investigation of the influence of fluorine on the supramolecular architecture of molecular crystals.

We consider two approaches for a description of crystal structures. Desiraju mentioned the first possibility of interpretation of crystal packing is a geometrical one,[24] which is based on Kitaigorodskii's close packing principle,[25] Etter's rules for crystals containing hydrogen bonds,[26] and the supramolecular synthon concept,[27] which are the basis of crystal engineering. The latter approach includes the recognition of some strongly bonded motifs in the crystal structure (dimers, trimers, tetramers, chains, etc.) based on strong intermolecular interactions.

In the case of the absence of strong classic intermolecular interactions, such as strong hydrogen bonds, the description of crystal packing becomes much more difficult. A similar situation is observed in the case of molecules with several comparable interactions. In this case, it is difficult to find the basic motif of the crystal packing for such crystals. A suitable approach to understand the aggregation phenomena could be based on the analysis of energies of the involved weak intermolecular interactions. This may be performed by modern ab initio quantum chemical methods with inclusion of electron correlation, for example, using perturbation theory or coupled cluster method.

Recent applications of this approach for the analysis of the crystal structures[28−30] demonstrated that even in the case of the absence of strong intermolecular interactions, it is always possible to recognize infinite motifs of a crystal structure (layers, columns). This allows the determination of the main structural pattern of crystal organization and one to describe the crystal structure in an unambiguous way.



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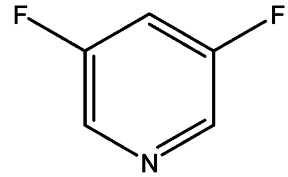

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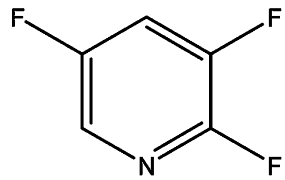

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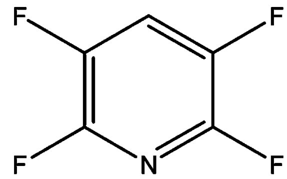

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In continuation of our investigations on aggregation phenomena,[18,31,32] and to understand how fluorine atoms influence the crystal packing in the absence of strong intermolecular interactions, we performed the first systematic study of a set of fluorinated pyridines, 3-fluoropyridine **1**, 3,5-difluoropyridine **2**, 2,3,5-trifluoro-pyridine **3**, 2,3,5,6-tetrafluoropyridine **4**, and pentafluoropyridine **V**. Compounds **1**−**4** were crystallized by in situ crystallization directly on the diffractometer. The structure of pentafluoropyridine **V** has been determined earlier.[19] The crystal structures have been analyzed using both geometrical and energetic approaches.

Table 1. Crystal Data and Details of Structure Refinement for 1−4

crystal data	1	2	3	4
formula	C_5H_4FN	$C_5H_3F_2N$	$C_5H_2F_3N$	C_5HF_4N
M_r/g mol^{-1}	97.09	115.08	133.08	151.07
crystal system	orthorhombic	monoclinic	triclinic	orthorhombic
space group	*Fdd*2	$P2_1$	$P\bar{1}$	*Pnma*
a/Å	23.44(4)	3.8400(8)	6.152(3)	6.773(4)
b/Å	21.79(4)	6.2700(13)	6.567(5)	7.654(4)
c/Å	3.776(6)	10.340(2)	7.250(5)	10.376(6)
α/deg	90	90	70.515(13)	90
β/deg	90	96.80(3)	75.307(14)	90
γ/deg	90	90	89.414(13)	90
V/Å^3	1928(6)	247.20(9)	266.2(3)	537.9(5)
Z	16	2	2	4
D_{calc}/mg m^{-3}	1.338	1.546	1.660	1.865
D_{calc}/ M_r/mmol cm^{-3}	13.78	13.43	12.47	12.35
μ/mm^{-1}	0.109	0.147	0.175	0.211
$F(000)$	800	116	132	296
T/K	173(2)	223(2)	198(2)	211(2)
T_{melt}/K	206.45	262.95	216.45	233
θ_{max}/deg	24.82	25.04	25.10	25.07
reflections collected/unique	1121/551	617/515	722/720	1313/439
data/parameters	551/74	515/74	720/91	439/50
S (F^2)	1.092	1.196	1.085	1.234
R_1/wR_2 ($I > 2\sigma$)	0.0769/0.1872	0.1080/0.2919	0.0816/0.2185	0.0588/0.1884
R_1/wR_2 (all data)	0.1052/0.2158	0.1311/0.3132	0.0882/0.2402	0.0654/0.1951

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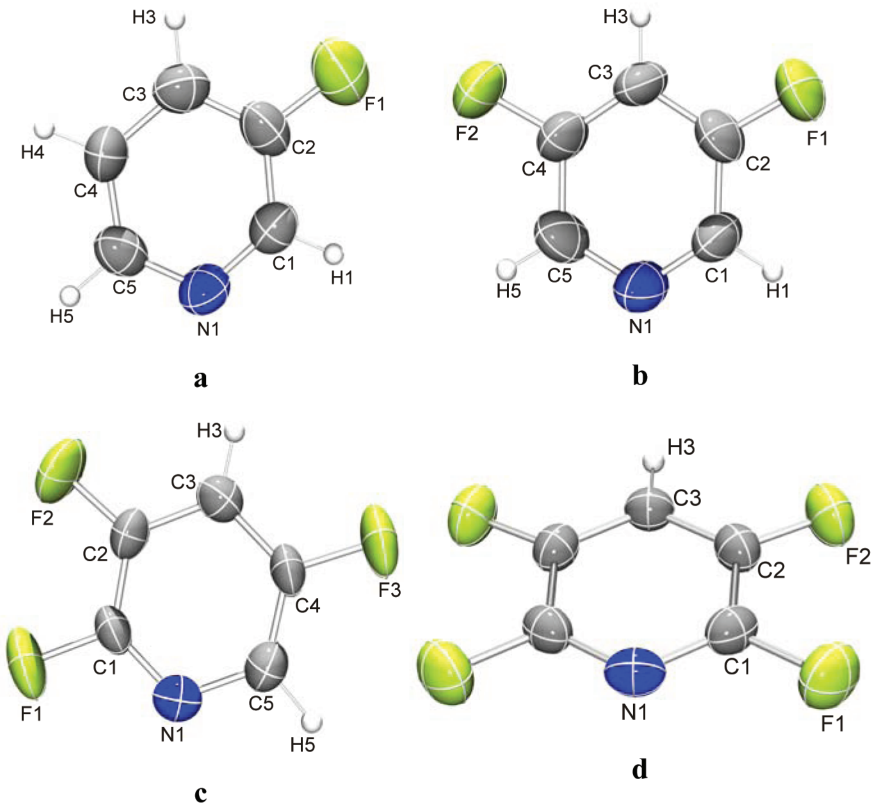

Figure 1. Molecular structures of (a) 3-fluoropyridine **1**, the disorder was excluded for simplification, (b) 3,5-difluoropyridine **2**, (c) 2,3,5-trifluoropyridine **3**, and (d) 2,3,5,6-tetrafluoropyridine **4**.

EXPERIMENTAL SECTION

Single-crystal X-ray diffraction measurements of **1**−**4** were carried out on a Bruker Smart 1000 CCD diffractometer using graphite-monochromated Mo Kα radiation. The compounds were first purified by distillation and then filled into a capillary. The capillary was sealed and transferred to the diffractometer, which is equipped with a cooling device. The single crystals suitable for X-ray diffraction were grown in situ using a computer-controlled device that applies a focused CO_2-laser beam along the capillary.[33] On the basis of the geometry of the low-temperature device, one omega-scan for each compound was collected. Structures were solved by the direct methods, and all non-hydrogen atoms were refined anisotropically on F^2 (program SHELXTL-97, G.M. Sheldrick, University of Goettingen, Goettingen, Germany).[34] All H atoms were detectable in difference maps. In **1**, **2**, and **4**, the H-atoms were positioned geometrically and refined using a riding model. For **1**, the fluorine atom was found to be disordered and therefore modeled in two equally occupied orientations. The crystallographic data and processing parameters are shown in Table 1.

Quantum-Chemical Calculations. The molecular structures of isolated fluorinated pyridines were fully optimized using the second order of the closed-shell restricted Møller−Plesset perturbation theory,[35] which has been widely used to study noncovalent interactions. All calculations were performed using standard 6-311G(d,p) basis set (MP2/6-311G(d,p) method). The presence of local minima on the potential energy surface was verified by establishing that the vibration frequencies have only positive values.

Taking into account the well-known effect of shortening lengths of the X−H bonds in the X-ray diffraction data,[36] these values were corrected according to the results of the geometry optimization of the isolated molecule of fluoropyridines. The corrected values of the C−H bond lengths of 1.08 and 1.09 Å, obtained from MP2/6-311G(d,p) calculations, were fixed without changes of corresponding bond and torsion angles.

To characterize the intermolecular interactions, the molecule in the crystallographically asymmetric unit (for $Z' \leq 1$), the basic molecule, was specified, and its first coordination sphere consisting of the neighboring molecules was determined. This was performed on the basis of calculations of the Voronoj−Dirichlet domains and coordination polyhedra of the basic molecule in the crystal structure.[37] Such polyhedra of the molecules were constructed as the sum of the Voronoj−Dirichlet polyhedron of all of the atoms of the molecule. Two molecules in the solid state are considered to be neighbors in the case when their polyhedra have at least one common face,[38] the set of which forms the boundary surface between two molecules. Fischer and Koch[39] consider the coordination number of a molecule in the crystal to be determined as a number of the boundary surfaces with each surface area exceeding 2% of a total area of the Voronoj−Dirichlet polyhedron of a molecule. Calculations of the Voronoj−Dirichlet polyhedron and their boundary surfaces are performed by the method by Panov et al.[40]

The basic molecule builds a dimer with each of the molecules in its first coordination sphere. For the calculation of the intermolecular interaction energy in each dimer, the geometry of all atoms was taken from the experimental data with corrected lengths of the C−H bonds (1.09 Å in the case of **1** and 1.08 Å for **2**, **3**, and **4**). The intermolecular interaction energy between molecules in each dimer of fluoropyridines was calculated by single point calculations using the MP2/6-311G(d,p) method as a difference between the energy of dimer and energy of the monomers, and it was corrected for a basis set superposition error using the standard Boys−Bernardi counterpoise procedure.[41] The total energy of interactions of the basic molecule with its environment in the crystal structure was determined as the sum of the intermolecular interaction energies of the dimers. All calculations were performed using the Gaussian 03[42] program.

RESULT AND DISCUSSION

Crystal Structures of 1−4 from the Geometrical Perspective. The replacement of hydrogen by fluorine is often considered as an isosteric substitution,[43] especially for monosubstituted analogues, as the fluorine substituted

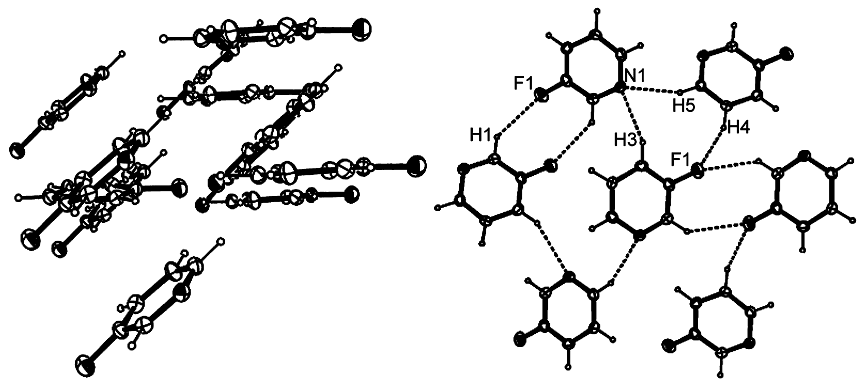

Figure 2. Crystal structure of **1** with selected interatomic interactions: C1−H1···F1 (2.55 Å), C5−H5···N1 (2.64 Å), C3−H3···N1 (2.61 Å), and C4−H4···F1 (2.55 Å). The disorder was excluded for simplification.

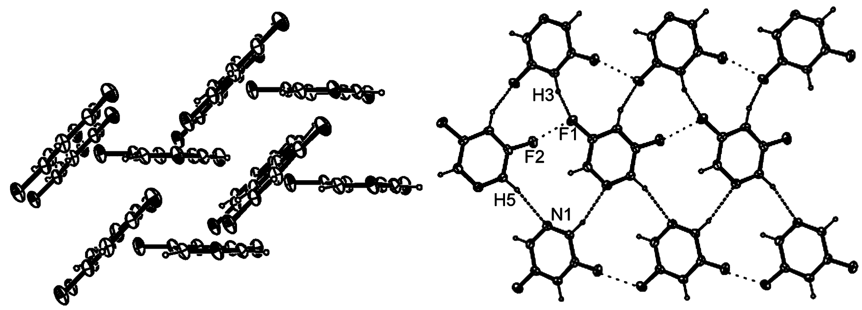

Figure 3. Crystal structure of **2** with selected interatomic interactions: C3−H3···F1 (2.62 Å), C5−H5···N1 (2.72 Å), and F1···F2 (2.93 Å).

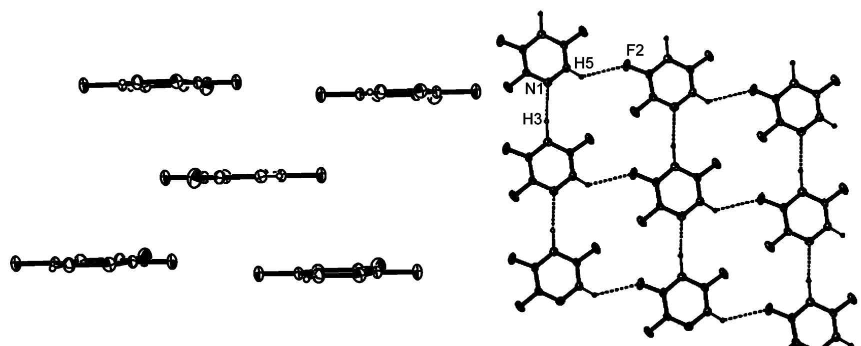

Figure 4. Crystal structure of **3** with selected interatomic interactions: C3−H3···N1 (2.20 Å) and C5−H5···F2 (2.64 Å).

compounds are geometrically very similar to the parent molecule. Pyridine crystallizes in the space group $Pna2_1$ with four molecules in the asymmetric part of the unit cell and contains layers perpendicular to the *c*-axis direction.[44]

In the case of 3-fluoropyridine **1**, the substitution of one hydrogen by a fluorine atom at the pyridine backbone leads to a rearrangement of the molecules. Compound **1** crystallizes in the orthorhombic space group *Fdd*2 (Figure 1a) and shows a positional disorder of the fluorine substituent at the 3 and 5 positions. The fluorinated pyridine **1** has a molecular C_s symmetry, rendered by the space group. The molecules form a herringbone arrangement with a distorted edge-to-face configuration. The angle between planes of neighboring molecules is 40.3°. In contrast, we had recently reported that the T-shape motifs in the crystal structure of 2-fluoropyridine show a reduced torsion angle of 58.8°.[18] Different C−H···N and C−H···F interactions were found in the crystal structure of **1** (Figure 2). Each nitrogen atom employs two C−H···N interactions (2.61 Å C3−H3···N1 and 2.64 Å C5−H5···N1) to two neighboring molecules. The C1−H1···F1 and C4−H4···F1 distances of 2.55 Å are below the sum of van der Waals radii of F···H (2.67 Å). Similar motifs of such C−H···F hydrogen bonds were found in the crystal structure of some fluorinated benzenes.[7]

3,5-Difluoropyridine **2** (Figure 1b) crystallizes in the monoclinic space group $P2_1$. The analysis of the crystal structure shows a distorted edge-to-face arrangement, similar to the crystal structure in earlier reported 2-fluoropyridine.[18] In comparison to 3-fluoropyridine, the angle between planes of neighboring molecules in **2** is slightly increased to 47.5°. Possible intermolecular interactions in the crystal structure of **2** are the C−H···N and C−H···F hydrogen bonds as well as the F···F interactions. The C5−H5···N1 interactions (2.72 Å) form zigzag chains (Figure 3). Each fluorine forms the F1···F2 contact of 2.93 Å, linking molecules in rows. The rows are arranged by the C3−H3···F1 interactions (2.62 Å).

2,3,5-Trifluoropyridine **3** (Figure 1c) crystallizes in the triclinic $P\overline{1}$ space group. The substitution of three fluorine

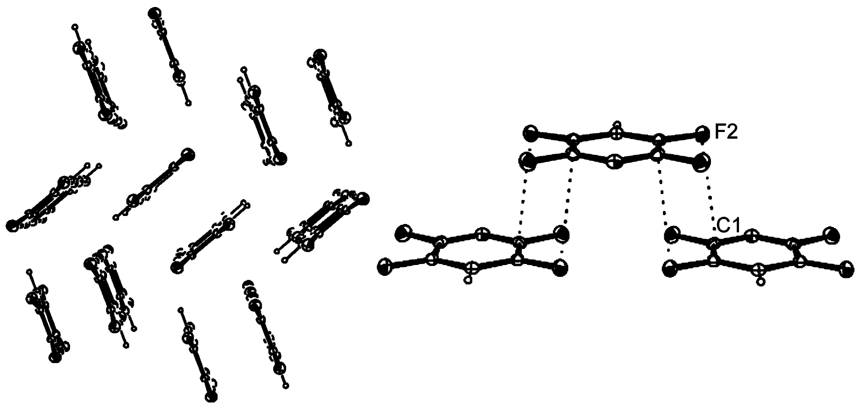

Figure 5. Crystal structure of **4** with selected interatomic interactions: C1···F2 (3.16 Å).

atoms on the pyridine backbone of **3** leads to a significant change of the character of the aggregation. The molecules are arranged in parallel and form stacking layers. The analysis of the crystal structure reveals different types of weak intermolecular interactions (Figure 4). Within the layer, the C5−H5···F2 (2.64 Å) and C3−H3···N1 (2.20 Å) interactions link molecules in chains along axes *c* and *a*, respectively, forming a two-dimensional network. The layers themselves are formed by stacking, consisting of $\pi\cdots\pi$ (3.34 Å), C1−F1···C1(π) (3.08 Å), and F1···F1 (2.92 Å) interactions.

2,3,5,6-Tetrafluoropyridine **4** (Figure 1d) aggregates in the orthorhombic *Pnma* space group with four molecules in the unit cell. A comparison with **3** shows that addition of one further fluorine atom in **4** leads to a sandwich herringbone packing of molecules, very similar to pentafluorobenzene[19] and the monoclinic polymorph of 1,2,3,4-tetrafluorobenzene.[45] In contrast to these structures, however, no C−H···F interactions are observed in the crystal structure of **4**. Beside the formation of the C−H···N hydrogen bond to the acidic H(1) in the *para*-position to N, each molecule in **4** forms four C2−F2···C1(π) interactions (3.16 Å), building stacking dimers as shown in Figure 5. However, the molecules in such stacked dimers cannot be strongly bonded because of the significant shift with respect to each other (Figure 5).

As expected, the analysis of the crystal structures of **1**−**4** reveals that there are no strong specific intermolecular interactions. The molecules in the solid state are linked by weak C−H···X (X = N, F) hydrogen bonds, $\pi\cdots\pi$ stacking, C−F···π, and F···F interactions.

To analyze the influence of the increase of fluorine substitution on the aggregation behavior of the pyridines, the crystal structures of **1**−**4** should be compared to the already known structure of pentafluoropyridine **V**. This fully fluorinated compound was investigated at high pressure by the group of Katrusiak.[19] The molecules of **V** aggregate in a herringbone pattern (Figure 6). The crystal structure displays different types of weak intermolecular interactions such as F···F (2.85 and 2.91 Å), C−F···π (3.08 Å), and N···π (3.12 Å).

Interestingly, we observe a stepwise alteration of the crystal packing with a growing number of fluorine substituents on the pyridine backbone, starting with edge-to-face aggregation in **1** and **2**, then $\pi\cdots\pi$ stacking in **3**, finally adapting a sandwich herringbone packing in **4**. Complete fluorination leads to the disappearance of $\pi\cdots\pi$ stacking interactions in the crystal structure of pentafluoropyridine and hence again to the edge-to-face packing. Recently, we reported a similar planarization

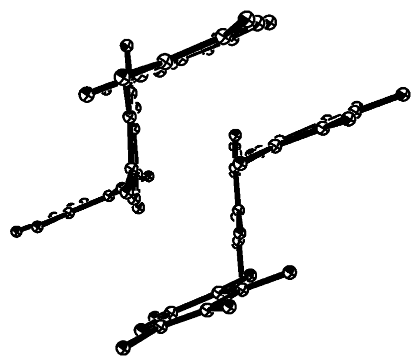

Figure 6. Crystal packing of **V** in the edge-to-face arrangement.

effect in the series 2-fluoro-, 2,6-difluoro-, and 2,4,6-trifluoropyridine.[18]

Crystal Packing Analysis from the Perspective of Intermolecular Interaction Energies. In general, the absence of strong intermolecular interactions complicates the description of the crystal structures of **1**−**4** and **V**. It is difficult to define the main structural motif in the solid state just on the basis of geometrical considerations. For instance, stacking interactions in **3** along the *b* axis are observed, whereas there are C−H···F interactions along the *c* axis and C−H···N hydrogen bonds along the axis *a*. It is not clear which interaction has the energetically dominating role in the crystal structure.

Therefore, an analysis of the energy of intermolecular interactions in the crystal structures **1**−**4** and **V** may provide more information about the organization of these crystals. The results of the Voronoj−Dirichlet polyhedra calculations show that the first coordination sphere of the basic molecule located in the asymmetric part of the unit cell of the crystal structure of **1**-**V** contains 12−14 neighboring molecules. Consequently, the respective number of dimers formed by the basic molecule and its closest neighbors was constructed for the further analysis (see the Supporting Information, Tables S1−S5).

The analysis of the intermolecular interaction energy for every dimer (Tables S1−S5) indicates the absence of strong interactions. The most attractive intermolecular interactions for dimers are presented in Table 2. The energy of the intermolecular interactions does not exceed −3.4 kcal/mol for all crystal structures. The most strongly bonded dimers contain stacking interactions, C−H···N hydrogen bonds, and N···π interactions.

Table 2. Intermolecular Interactions with the Highest Energies for Dimers in Crystal Structures of 1−4 and V

compound	dimer no.	operations of symmetry	$-E_{int}^{dimer}$, kcal mol^{-1}	primary interaction	contacts, Å
1	3, 4	$x + 0.25, -y + 0.25, z + 0.25; x - 0.25, -y + 0.25, z - 0.25$	2.50	C3−H···N1′	2.504
	1, 2	$x, y, z - 1; x, y, z + 1$	2.50	$\pi\cdots\pi'$	3.776
	9, 10	$-x + 0.5, -y, z - 0.5; -x + 0.5, -y, z + 0.5$	2.32	C5−H···N1′	2.526
2	1, 2	$x - 1, y, z; x + 1, y, z$	2.85	$\pi\cdots\pi'$	3.840
	7, 8	$-x, y - 0.5, -z; -x, y + 0.5, -z$	2.32	C1−H···N1′	2.756
	11, 12	$-x - 1, y - 0.5, -z; -x - 1, y + 0.5, -z$	1.50	C5−H···N1′	2.583
3	2	$-x, -y + 1, -z + 1$	3.31	C1···C1′ ($\pi\cdots\pi'$)	3.344
	5, 6	$x - 1, y, z; x + 1, y, z$	2.97	C3−H···N1′	2.304
	1	$-x - 1, -y, -z + 2$	1.84	C4···C4′ ($\pi\cdots\pi'$)	3.339
4	1, 2	$x - 0.5, -y + 0.5, -z + 0.5; x + 0.5, -y + 0.5, -z + 0.5$	3.17	N1···π	3.211
V	2, 3	$-x + 1, y - 0.5, -z + 1.5; -x + 1, y + 0.5, -z + 1.5$	2.31	N1···C5′ (π)	3.117
	4, 5	$x, y - 1, z; x, y + 1, z$	2.04	$\pi\cdots\pi'$	5.179

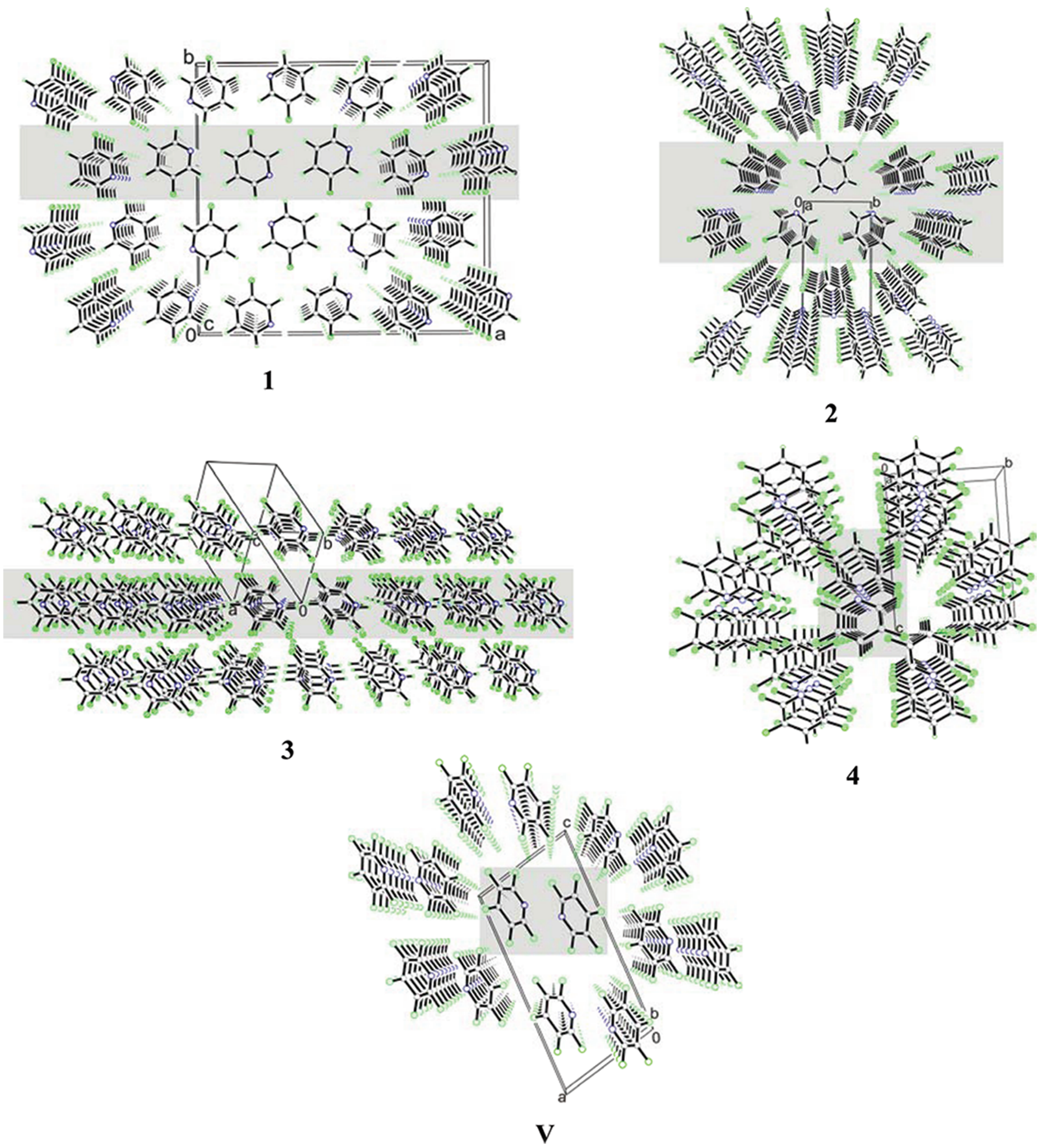

Figure 7. Crystal packing of **1**−**4** and **V** from the energetical perspective. BSMs, constructed on the basis of the calculated intermolecular interaction energies, are highlighted in gray: **1**, layer; **2**, double layer; **3**, layer; **4**, double column; **V**, double column.

The results of our calculations demonstrate that the dimers containing C−H···F interactions in the crystal structure of **1**−**4** with the energy values between −1.14 and −0.40 kcal mol^{-1} (Tables S1−S4) do not have a dominating role in the formation of crystal structures. Just slightly negative values of −0.04 and −0.07 kcal mol^{-1} (Tables S2, S3) and even a positive value (+0.09 kcal/mol; Tables S4, S5) were obtained in the case of dimers involving intermolecular F···F interactions. This indicates a very weak attractive or rather repulsive nature of fluorine−fluorine interaction.

Taking into account only the dimers with the highest energies of intermolecular interactions, it is possible to reconstruct the crystal architecture of the analyzed compounds and determine the main structural motif in a crystal structure by using the procedure described recently.[28−30] The dimers with the strongest interactions determine the strongly bonded energetically rich fragment of the crystal, the basic structural motif (BSM). This fragment also includes several weaker bonded dimers. Therefore, the crystal structure consists of the infinitely recurring BSMs. The energy of the interactions within the BSM is calculated as a sum of the energies of the dimers belonging to this BSM. The energy of the interactions between two neighboring BSMs is determined as a sum of the interaction energies of the basic molecule and the molecules of its first coordination sphere belonging to the BSM other than the basic molecule.

The results of the calculations demonstrate that the total energy of intermolecular interactions within one BSM is significantly higher as compared to the total energy of interactions between two neighboring BSMs, so that energetically rich BSMs and energetically poor inter-BSM spaces of the crystals can be defined.

In the crystal structure of **1**, the C−H···N interactions (dimers 3 and 4, Table 2) link molecules producing chains, and $\pi\cdots\pi$ interactions (dimers 1 and 2) form columns so that a two-dimensional network, a layered structure is realized, based on the dimers with the highest intermolecular interaction energies. Figure 7, **1** represents the BSM in **1**, layer (highlighted in gray) parallel to the (001) crystallographic plane. The construction of the BSM showed that dimers 9 and 10 formed by the C5−H···N1′ interaction do not belong to the same layer as the basic molecule. These hydrogen bonds correspond to the interactions between two BSMs and stabilize herewith the inter-BSM space. A similar crystal organization is found for the crystal structure of **3** with a layer as BSM. The crystal structure of **2** reveals a double layered structure. Both structures, **2** and **3**, are formed similar to the crystal structure of **1** by C−H···N and stacking interactions. Each layer/double layer has two equal neighboring layers/double layers, as shown in Figure 7, **1**−**3**. The crystal packing of **4** is dominated by N···π interaction, whereas **V** includes N···π and stacking interactions, forming both double columned structures. The main double columns are surrounded by six neighbors.

Whereas for **1**−**3** and **V** the intermolecular interactions energies are equal to each of their neighboring BSMs, in structure **4** a stronger interaction with four columns along axis *c* (−2.46 kcal/mol) is observed. The energy of these interactions is still too weak in comparison with the energy within the double column (Table 3), so that the crystal cannot be considered to have a layered structure. The interactions with further two BSMs along *b* are slightly repulsive with +0.09 kcal/mol. The geometrical orientation of these double columns corresponds to the F···F interactions.

As mentioned above, the dimers based on the F···F interactions show weak attractive, sometimes even slightly repulsive, intermolecular interactions. This result indicates that the F···F interaction is formed just by close packing and is not directly participating in the composition of the crystal packing. Nevertheless, the F···F interactions are constructive, but this is

Table 3. Comparison of the Intermolecular Interaction Energy within the Basic Structural Motif (E_{BSM}) with the Energy between Two Neighboring Motifs (E_{ext})

compound	basic structural motive	E_{BSM}, kcal mol^{-1}	E_{ext}, kcal mol^{-1}
1	layer	−11.23	−4.17
2	double layer	−14.52	−1.95
3	layer	−13.45	−2.62
4	double column	−9.51	−2.46
V	double column	−8.71	−0.93

mostly based on dispersion or/and on the electrostatic interactions when the fluorine atoms are polarized in a different manner. In the series of the crystal structures of **1**-**V**, fluorine definitely has an influence on the aggregation of the molecules. The resulting BSMs (Figure 7) show that the molecules in all analyzed structures are orientated in such a way to achieve the farthest possible distance between the fluorine atoms. This is especially strongly pronounced on the packing of **2** and **V** (Figure 7). F···F interactions do not participate in the formation of the strongly bonded energetically rich fragment of the crystals. These interactions are only observed in the energetically poor inter-BSM spaces.

The results show that it is possible, based on the calculations of the intermolecular interaction energy between the basic molecule and each of the molecules in its first coordination sphere, to recognize the real basic structural motif in the solid state. The main crystal motifs for the analyzed compounds include layers, double layer, and double columns of strongly interacting molecules linked by considerably weaker C−H···F, F···F, and C−H···π interactions.

The character of BSM depends on the number and location of fluorine atoms. The presence of the C−H group in *ortho*- and *para*-position with respect to the nitrogen atom of pyridine ring results in formation of the layered structures with single (structures **1** and **3**) or double (structure **2**) layers as BSM. In the case of further substitution of hydrogen atoms by fluorine, BSM is transformed to double columns (structures **4** and **V**). These results show especially the importance of the C1−H···N hydrogen bonds for organization of fluoropyridine crystals. In the case of absence of a possibility for the formation of such hydrogen bonds, the N···π interactions become the most important, causing the formation of columnar structures.

■ CONCLUSION

The results of the first systematic study of the crystal structures of fluorinated pyridines reveal consecutive changes of character in the crystal organization with increasing number of fluorine atoms. The main feature of these crystals is the absence of any strong intermolecular interactions leading to problems with the determination of supramolecular synthons. Therefore, the analysis of the supramolecular architecture of the crystals was performed using both a traditional geometrical approach and a method based on the consideration of the values of energy of intermolecular interactions between the basic molecule located in the asymmetric part of a unit cell and neighboring molecules belonging to its first coordination sphere.

According to geometrical considerations, the crystal packing of 3-fluoropyridine **1**, 3,5-difluoropyridine **2**, 2,3,5-trifluoropyridine **3**, and 2,3,5,6-tetrafluoropyridine **4** changes stepwise in dependence of the number of fluorine atoms. Going from the monosubstituted to the perfluorinated compound, pentafluoropyridine **V**, the crystal packing alters stepwise from herringbone packing to parallel arrangement of the molecules in trifluorinated pyridine and switches again to the edge-to-face form.

A deeper insight into the crystal organization is obtained from the analysis of the energies of intermolecular interactions calculated by the MP2/6-311G(d,p) method. Two dimers that are formed by the basic molecule and a molecule from its first coordination sphere, with the highest value of interaction energy, determine an infinite motif of the crystal structure called basic structural motif, BSM. The energy of the interactions of the basic molecule with molecules belonging to BSM is 3−7 times higher than the energy of interaction of the basic molecule to molecules belonging to neighboring BSM. Therefore, the crystal structures may be presented as a packing of BSMs containing strongly interacting molecules.

Taking into account the structure of BSM, one can conclude that intermolecular interactions involving fluorine atoms are not so important for the crystal organization. However, a detailed analysis of BSMs indicates that in all cases the fluorine atoms are oriented mainly out of the BSM providing maximal F···F distances. Therefore, it is likely that the presence of fluorine atoms also considerably influences the crystal packing. The molecules are arranged in such a way to minimize the F···F repulsion within strongly bonded BSM. The interactions involving fluorine atoms are mainly responsible for the interaction between neighboring BSMs.

■ ASSOCIATED CONTENT

S Supporting Information

Values of the intermolecular interaction energies in dimers of the crystal structure of **1-V** are listed in Tables S1−S5. CIF files giving X-ray data with details of refinement procedures for 3-fluoropyridine, 3,5-difluoropyridine, 2,3,5-trifluoropyridine, and 2,3,5,6-tetrafluoropyridine (CCDC 835253−835256). This material is available free of charge via the Internet at http://pubs.acs.org.

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